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ELEC5870M Final Report

Fabrication of Enhancement Mode HEMT Devices via Gate Recession

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Abstract

Abstract

This report documents the complete ELEC5870M MEng project on fabricating enhancement-mode GaAs/AlGaAs high-electron-mobility transistors via controlled gate recession. It will first summarise the validated baseline process and depletion-mode electrical benchmarks established during Semester 1 before detailing the Semester 2 process innovations that enabled tightly-controlled recess depths. The final chapters will correlate recess depth with threshold voltage, transconductance, and gate leakage to demonstrate the tuning range achieved, and will reflect on the manufacturability of the proposed flow.

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Abbreviations

2DEG	Two-dimensional electron gas
B1500A	Semiconductor parameter analyser
DI	Deionised water
HEMT	High-electron-mobility transistor
HCl	Hydrochloric acid
III-V	Group III-V semiconductor
IPA	Isopropanol
LOR	Lift-off resist
MLA	Maskless aligner
MMIC	Monolithic microwave integrated circuit
RTA	Rapid thermal annealing
SEM	Scanning electron microscopy

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CHAPTER 1

Introduction

This chapter will concisely describe the motivation for normally-off III–V transistors, present the specific objectives of fabricating and benchmarking recessed-gate GaAs/AlGaAs HEMTs, and highlight the dual-semester workflow used in this project. Content migrated from the interim report will be reframed to emphasise how the validated depletion-mode process underpins the enhancement-mode experiments.

1.1 Motivation for Enhancement-Mode GaAs HEMTs

Future iterations should summarise the limitations of depletion-mode devices for RF and power electronics, including the need for negative gate drive rails and shoot-through risks, and explain why GaAs remains a useful research platform despite the popularity of GaN.

1.2 Project Objectives and Scope

This section will define the quantitative goals for threshold voltage shift, process reproducibility, and electrical benchmarking. It will also introduce the comparative methodology between baseline and recessed devices.

1.3 Report Organisation

A short roadmap will link each subsequent chapter to the overarching research questions, enabling readers to navigate fabrication, characterisation, and project management content efficiently.

CHAPTER 2

Technical Background

This chapter will consolidate the theoretical material that links heterostructure design, carrier transport, and threshold voltage control. Existing text from the interim report can be merged with new literature covering oxide removal, alternative recess approaches, and reliability considerations.

2.1 GaAs/AlGaAs Heterostructure Overview

Outline the epitaxial stack, highlighting how the cap, donor, spacer, and channel layers determine sheet charge density and gate control.

2.2 Charge Control and Threshold Voltage Models

Discuss parallel-plate approximations, quantum corrections, and the analytic relationship between donor thickness and threshold voltage. This section sets the theoretical expectations for the recess experiments.

2.3 Gate Recess Strategies in Literature

Compare wet versus dry etching, oxide mitigation strategies, and prior reports of enhancement-mode GaAs HEMTs to justify the selected process window.

2.4 Measurement Fundamentals

Summarise the electrical metrics of interest (ID–VDS, ID–VG, gm, leakage) and how they relate to the physics described above.

CHAPTER 3

Baseline Process Development

This chapter will expand on the Semester 1 fabrication work, documenting all process parameters needed to reproduce the depletion-mode GaAs/AlGaAs HEMT array. The narrative should emphasise learning outcomes, yield, and open issues that informed Semester 2 planning.

3.1 Mesa Definition and Isolation

Include lithography settings, etch chemistry, profilometry data, and lessons on etch-rate discrepancies.

3.2 Ohmic Contact Formation

Detail the dual-layer resist scheme, metallisation stack, RTA profile, and contact resistance characterisation.

3.3 Gate Metallisation

Summarise alignment strategy, Ti/Au deposition, lift-off cleaning, and quality control images.

3.4 Baseline Device Inventory

Provide tables of geometries, wafer layout diagrams, and discussion of process repeatability across the fifteen devices.

CHAPTER 4

Baseline Electrical Characterisation

Here we will collate all unrecessed device measurements, focusing on the metrics that will later be compared against recessed devices. Figures and statistics from the interim report can be updated with any refined analysis.

4.1 Measurement Setup

Describe the B1500A configuration, probe station workflow, temperature controls, and data-processing scripts.

4.2 Output Characteristics

Summarise representative ID–VDS curves, contact resistance extraction, and implications for process quality.

4.3 Transfer Characteristics and Threshold Extraction

Document VT extraction methods, gm behaviour, and variability across device geometries.

4.4 Gate Leakage Behaviour

Provide leakage trends, discuss interface quality, and establish the reference point for recessed devices.

CHAPTER 5

Gate Recess Process Engineering

This chapter will describe the Semester 2 process innovations that enabled precise gate recessing, including calibration structures, etch timing models, and oxide management.

5.1 Process Objectives

Define the targeted recess depths, allowable tolerances, and link to the threshold-voltage goals set in Chapter 1.

5.2 Etch Chemistry Selection and Calibration

Report measured etch rates, agitation schemes, and profilometry data for the $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2:\text{H}_2\text{O}$ solution. Include lessons from calibration chips.

5.3 Native Oxide Mitigation

Detail the HCl dip procedure, timing constraints, and any surface-analysis results.

5.4 Mask and Layout Strategy

Explain the quadrant-based recess matrix, sample labelling convention, and any cleanroom logistics considerations.

CHAPTER 6

Enhancement-Mode Iteration 1: Fabrication and Results

6.1 Fabrication Process

6.2 Electrical Results

6.3 Analysis and Diagnosis

CHAPTER 7

Enhancement-Mode Iteration 2: Revised Mask and Results

7.1 Fabrication Process

7.2 Electrical Results

7.3 Analysis and Diagnosis

CHAPTER 8

Enhancement-Mode Iteration 3: Refined Process and Results

8.1 Fabrication Process

8.2 Electrical Results

8.3 Analysis

CHAPTER 9

Discussion and Failure Analysis

9.1 Summary of Experimental Outcomes

9.2 Failure Analysis

9.3 Process Limitations

9.4 Project Management Reflection

CHAPTER 10

Conclusions and Future Work

10.1 Conclusions

10.2 Future Work

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APPENDIX A

Supplementary Data and Process Logs

This appendix will gather supporting material such as extended etch-rate measurements, cleanroom log excerpts, supplementary microscope images, and code listings referenced throughout the report. Populate the subsections below as data becomes available.

A.1 Etch Calibration Tables

A.2 Cleanroom Session Notes

A.3 Additional Figures

A.4 Data Processing Scripts